

Tristan Toshiharu Endo

✉ ttendo@ucsd.edu ☎ +1-415-730-8521 in linkedin.com/in/tristan-endo 📄 github.com/Tristane028

Education

University of California, San DiegoSept 2024 – Jun 2027
B.S. Math-Computer Science — GPA: 3.52(Expected)

- Relevant Coursework: Real Analysis, Stochastic Processes, Brownian Motion, Mathematics in Finance, Enumerative Combinatorics, Probability Theory, Abstract Algebra

Research Experience

Undergraduate Researcher — Advisor: Alexander CloningerJan 2026 – Mar 2026
Neural Network Gaussian Processes & Conjugate Kernels

- Derived closed-form conjugate kernels for infinite-width ReLU and erf networks, confirming alignment between finite-width simulations and theoretical kernel recursion predictions
- Investigated kernel degeneracy, spectral bias, and expressivity as functions of activation choice and weight variance, comparing finite-width networks to their infinite-width Gaussian process limits

Undergraduate Researcher — Advisor: Raymond O. ChouFebruary 2026 – Present
Springer Fibers (Algebraic Geometry & Combinatorics)

- Investigate the geometry and combinatorics of Springer fibers, analyzing flag variety structure and symmetric group actions to characterize irreducible components

Industry Experience

Quantitative Researcher / AI Engineering InternJanuary 2026 – Present
Tekly Studio — Quantitative Research & AI Engineering LabSan Francisco, CA

- Developed and backtested ML-driven trading strategies on financial time series, evaluating performance via Sharpe ratio and drawdown analysis
- Engineered data pipelines for preprocessing, simulation, and model evaluation, enabling rapid iteration across strategy variants

Audio Engineer / AnalystApril 2025 – Present
Think Twice LegalSan Bruno, CA

- Optimized Demucs-based source separation models via hyperparameter tuning, improving reconstruction fidelity for forensic audio analysis in legal proceedings
- Built preprocessing and normalization pipelines for consistent model input across varying sample rates and noise conditions

Selected Projects

Stochastic Simulation of Brownian Motion & OU ProcessesSep 2025 – Dec 2025

- Simulated fractional Brownian motion across varying Hurst parameters to characterize long-range dependence and correlation structure
- Quantified empirical distributions of extrema via large-scale Monte Carlo simulation, validating convergence to theoretical predictions across parameter regimes

Audio Signal Processing & ML Pipeline DevelopmentApr 2025 – Present

- Designed FFT-based noise reduction models to isolate target signals from complex audio environments
- Built automated stem separation pipelines using Demucs with normalization workflows for consistent model input across varied audio conditions

Technologies

Languages: C, C++, Java, Matlab, Python, R, SQL

Libraries & Tools: PyTorch, TensorFlow, NumPy, SciPy, Torchaudio, FFmpeg, L^AT_EX, VS Code, RStudio, Tableau, SQL Workbench

Areas: Algorithmic Trading, Time-Series Modeling, Stochastic Processes, Audio Source Separation, Statistical Modeling